

Zerve Upgrade Prediction

Project Report from Notebook Analysis

Scope

This report summarizes the notebook work for the Zerve telemetry challenge: data understanding, leakage-safe feature engineering, upgrade prediction, funnel design, behavioral insights, and visual evidence. It intentionally excludes the interactive application layer.

- Dataset: 3,509,628 telemetry rows across 83 columns.
- Target: subscription_upgraded, modeled at the user level.
- Final feature table: 16,496 users and 43 predictive features.
- Model: XGBoost with stratified 5-fold cross-validation.
- Funnel: deterministic, mutually exclusive user stages.

Executive Summary

Objective

Predict which users are likely to upgrade using only pre-upgrade behavior, then translate model outputs into practical growth and product actions.

Core Method

The analysis builds a per-user snapshot time: first upgrade moment for upgraders and last observed event for non-upgraders. Features are computed only before that snapshot, with commercial and checkout-adjacent events excluded from the modeling pipeline.

Model Result

The XGBoost model achieved 0.7828 +/- 0.0156 ROC-AUC across 5 stratified folds, with 0.0928 +/- 0.0187 PR-AUC. The PR-AUC is the more realistic secondary metric because the target class is rare.

Product Result

Users are bucketed into Low, Moderate, High, and Very High intent tiers. The tiers attach model scores to recommended actions such as onboarding nudges, contextual prompts, personalized outreach, and priority sales contact.

Funnel Result

A deterministic funnel assigns every user to exactly one current stage: New, Exploring, Builder, AI User, Power User, Upgraded, or At Risk. The funnel uses observable event-count thresholds rather than model predictions.

Data Understanding and Leakage Controls

Dataset

The notebook analyzes zerve_events.csv, containing 3,509,628 rows and 83 columns of product telemetry. The raw event stream includes 227 distinct event types across pageviews, AI usage, canvas and block activity, deployments, credits, subscription actions, and person-level signup fields.

Missingness

Sixty of the 83 columns contain null values. This is expected because many fields are event-specific: AI token and latency fields populate only for AI generation events, while browser and GeoIP fields populate only for browser-sourced events.

Leakage Prevention

The notebook excludes direct target events and high-risk commercial signals from feature generation. It also cuts each user's history at the snapshot time so the model cannot learn from post-upgrade behavior.

Feature Table

After leakage filtering, 2,890,904 safe events remain. The final modeling table contains 16,496 users, 43 predictive features, and one label.

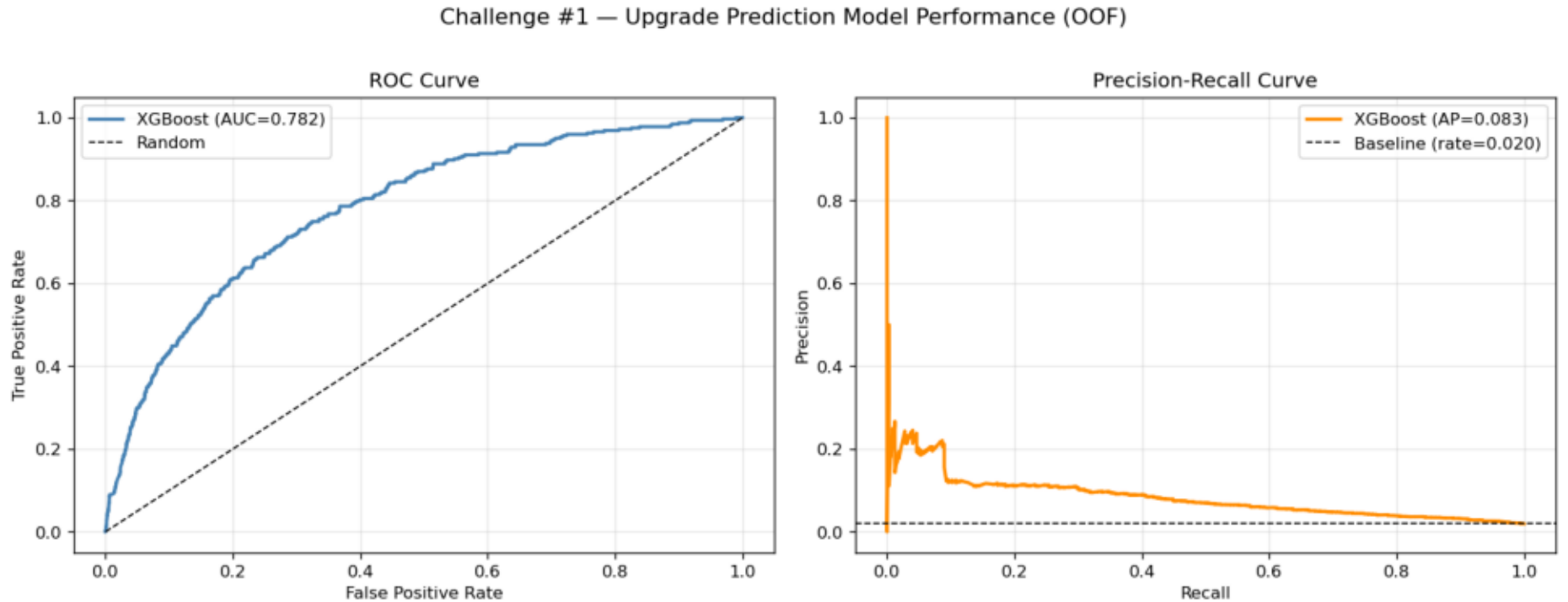
Modeling Features

Representative feature families created before each user snapshot.

Feature Family	Examples	Why It Matters
Engagement volume	total_events, active_days, session_count	Captures depth and consistency of product usage.
AI usage	ai_gen_count, ai_tokens_total, ai_latency	Measures activation around the product's high-value AI workflow.
Recent activity	events_last_7d, events_last_14d, pct_active	Separates active prospects from stale users.
Creation behavior	canvas_create_count, block_create_count	Signals movement from browsing to building.
Production intent	deployment_count, notebook_deploy_usage	Captures users moving toward operational use.
Credit pressure	credit_pressure_count, has_hit_credit_limit	Identifies quota moments that can motivate upgrade.

Model Performance

ROC and precision-recall performance for the leakage-safe XGBoost model.



What this output shows

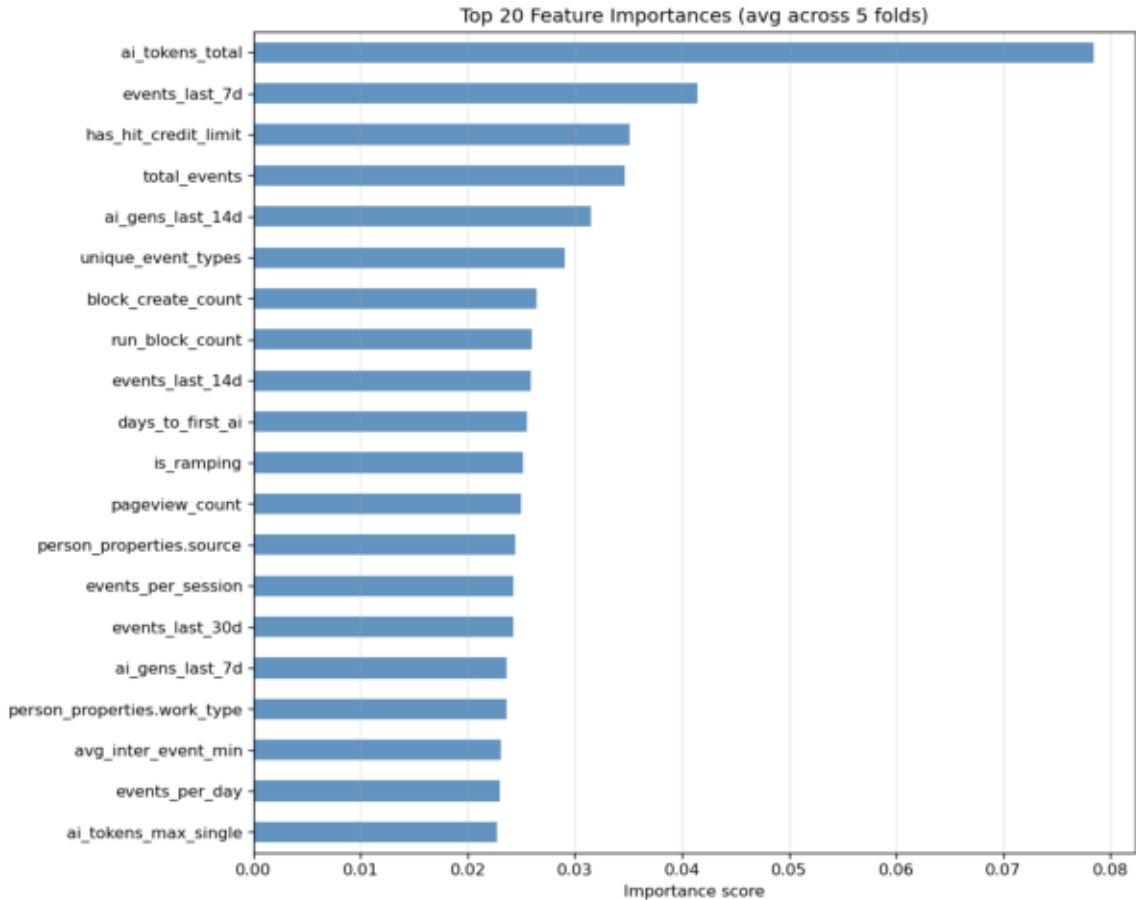
The ROC curve measures how well the classifier separates upgraders from non-upgraders across thresholds. The precision-recall curve focuses on positive-class retrieval, which is important because upgrades are rare.

What can be inferred

The model has meaningful rank-ordering power, but the low base rate means teams should use it for prioritization and tiering rather than treating each probability as a guaranteed outcome.

Feature Importance

Top model features emphasize AI usage, recent activity, credit pressure, and engagement depth.



What this output shows

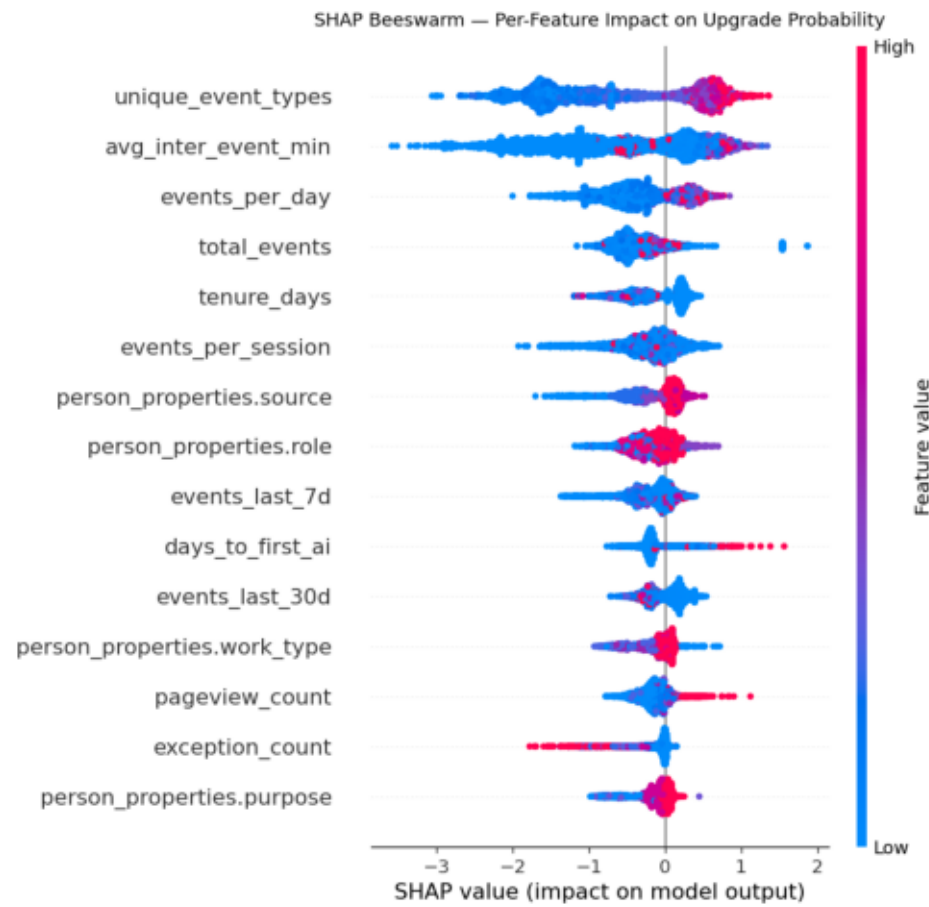
This chart ranks the features XGBoost used most heavily when splitting users into higher and lower upgrade-propensity groups.

What can be inferred

AI token volume, recent events, credit pressure, total activity, and breadth of event types are among the strongest signals, suggesting upgrade intent is tied to both value discovery and quota pressure.

SHAP Interpretability

Per-user SHAP effects show how feature values push upgrade probability up or down.



What this output shows

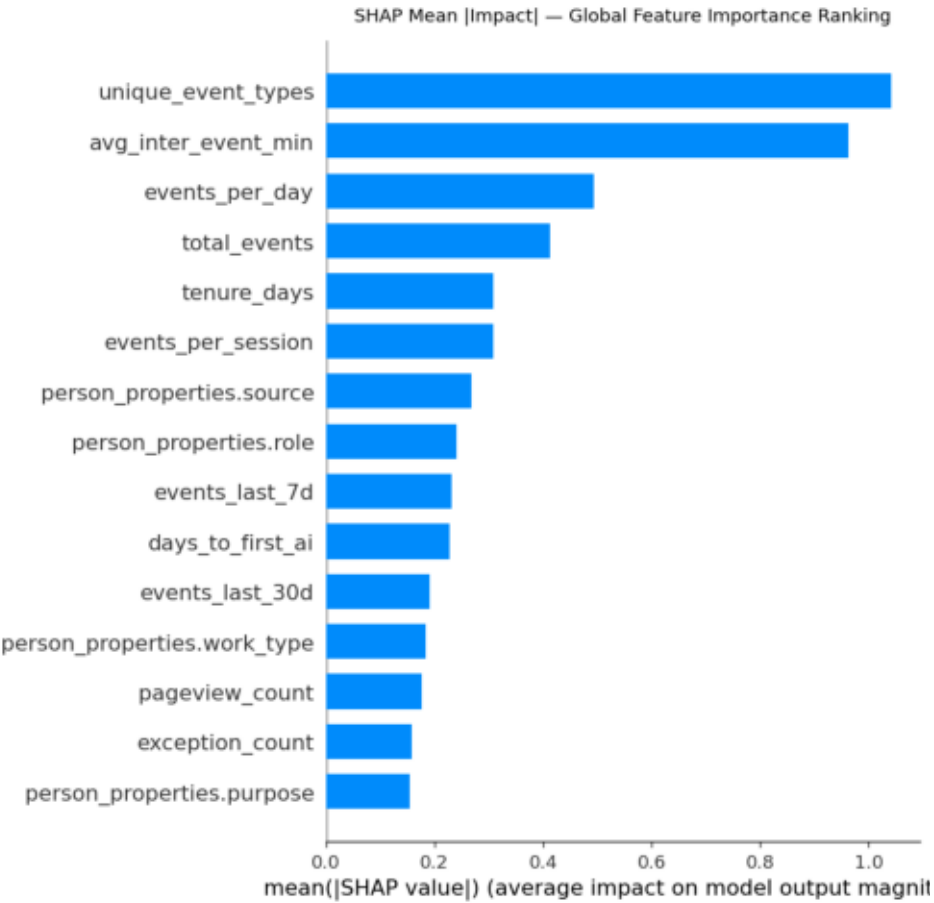
Each dot is a user-level contribution for one feature. Position shows whether the feature pushed the prediction higher or lower; color shows whether that user had a high or low value for the feature.

What can be inferred

High AI usage, recent activity, and credit-pressure related behavior generally push predictions upward, while low engagement patterns push users toward lower upgrade probability.

Global SHAP Ranking

Mean absolute SHAP values provide a global feature importance ranking.



What this output shows

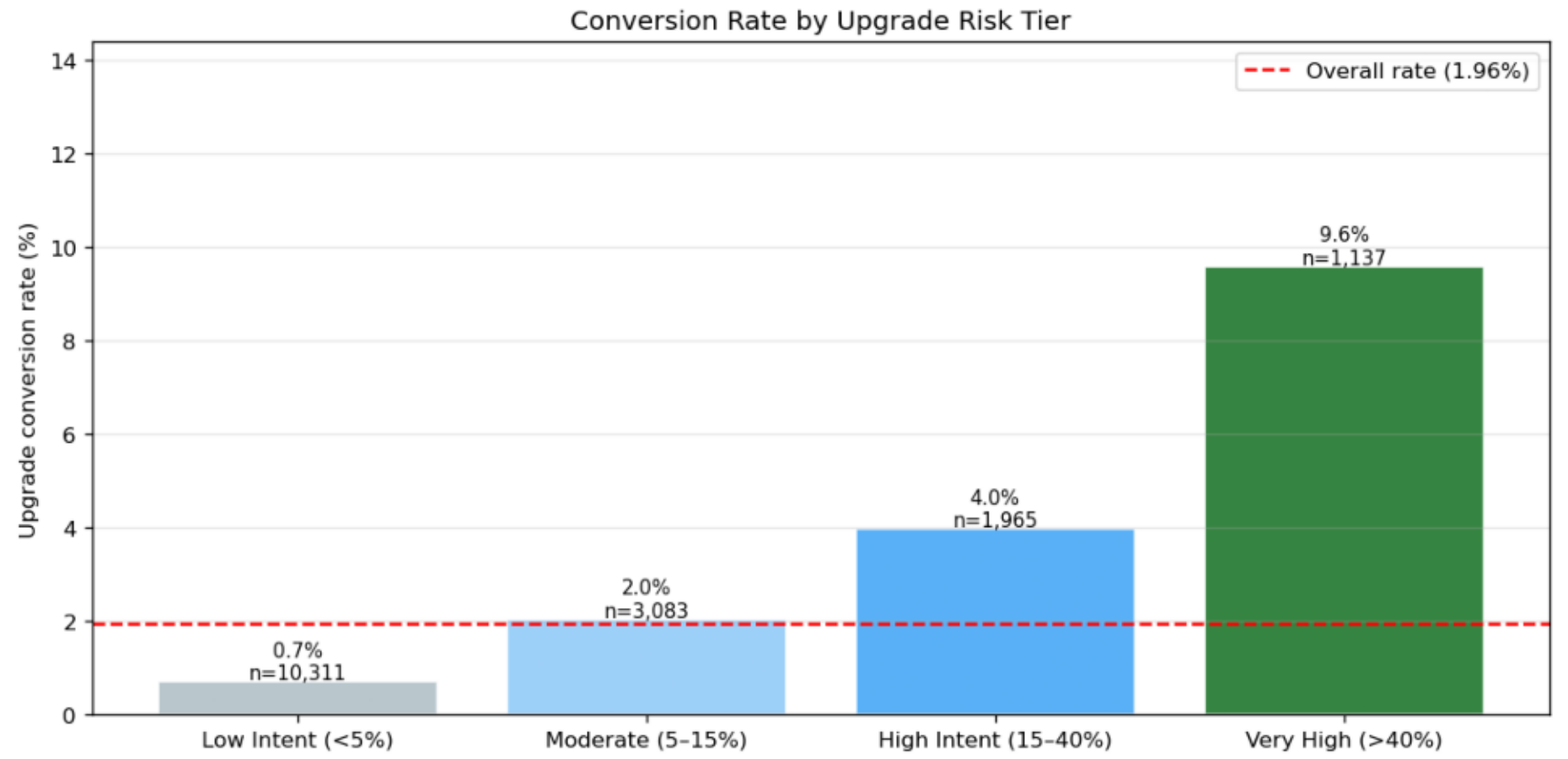
The bar chart aggregates absolute SHAP effects across users to show which features most consistently influence model predictions.

What can be inferred

The same behavioral themes dominate globally: AI usage, recent engagement, credit pressure, and creation/run activity. This supports a product strategy centered on activation depth.

Actionable Risk Tiers

Probability buckets convert raw model scores into growth team actions.



What this output shows

Users are grouped into Low, Moderate, High, and Very High intent tiers, with user counts, upgrader counts, conversion rates, and recommended actions.

What can be inferred

The Very High tier has the strongest conversion concentration, so it should receive scarce high-touch resources. Lower tiers are better suited to scalable nudges and onboarding.

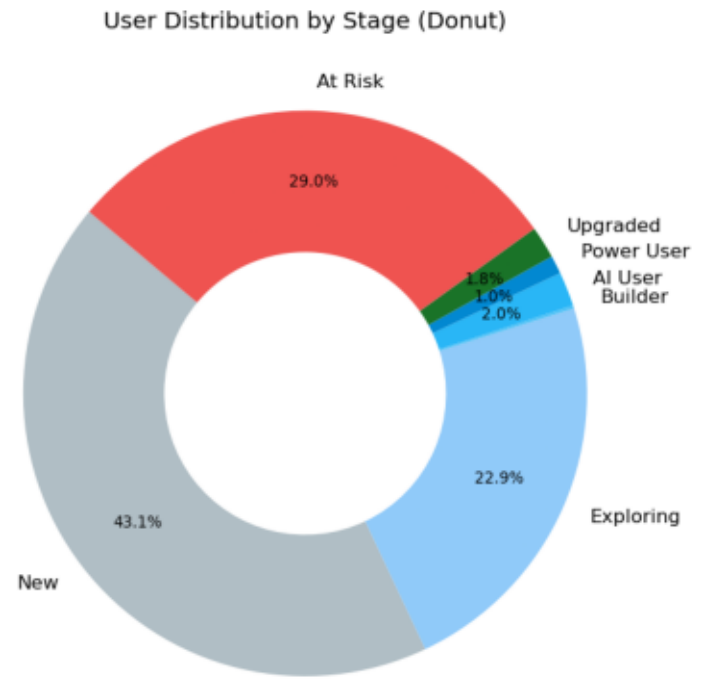
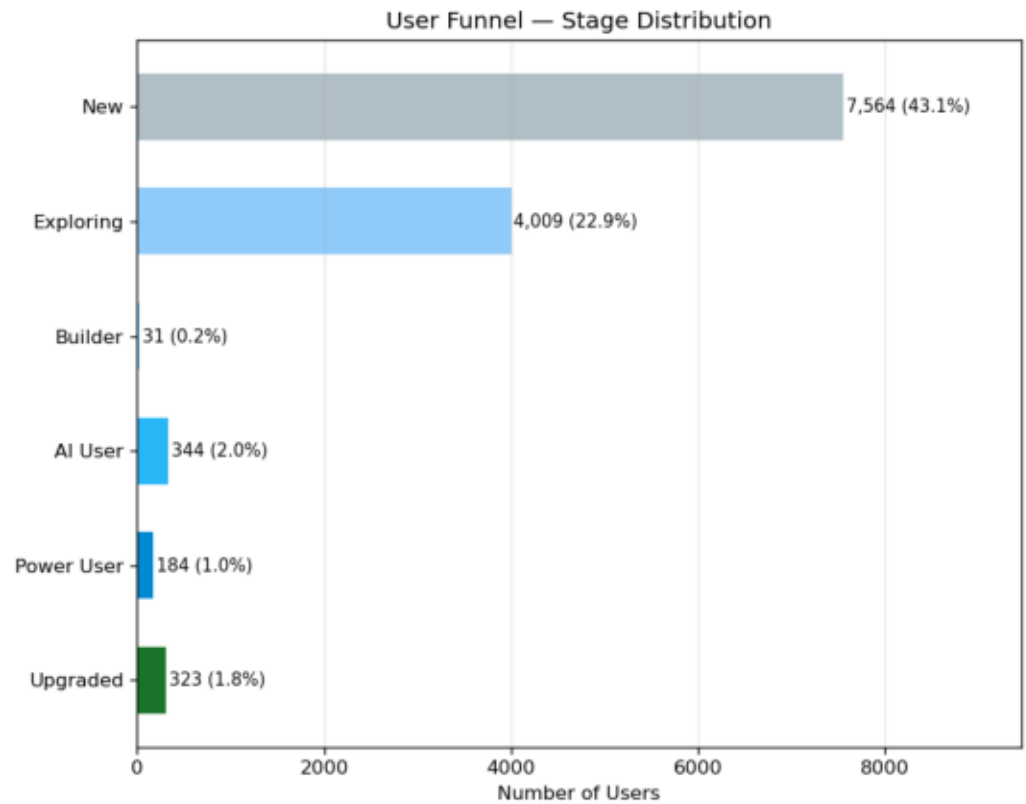
Funnel Definition

Deterministic stages based only on observable behavior.

Stage	Rule
New	Any event but no meaningful product interaction.
Exploring	At least one pageview.
Builder	Created at least one canvas or block.
AI User	Used AI at least once.
Power User	Deployed something or ran blocks at least five times.
Upgraded	Has a subscription_upgraded event.
At Risk	Was Builder or higher and inactive for 14+ days.

Funnel Distribution

Current-stage distribution across the user base.



What this output shows

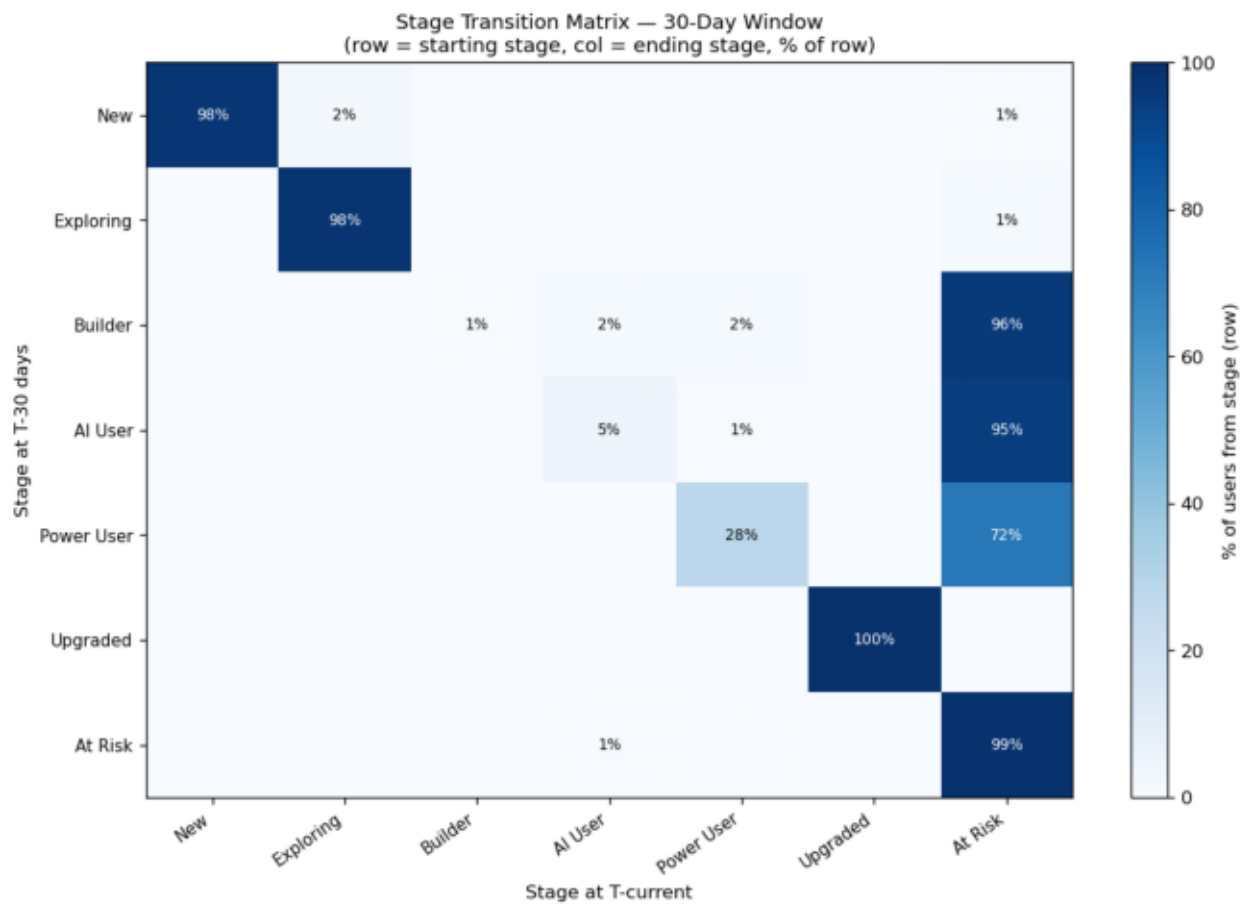
The funnel chart assigns every user to a deterministic stage based on observed product behavior, from New through Upgraded and At Risk.

What can be inferred

A large share of users remain New, Exploring, or At Risk, which points to two opportunities: improve early activation and re-engage previously meaningful users before they decay.

Stage Transition Matrix

Thirty-day movement validates the stage rules and progression logic.



What this output shows

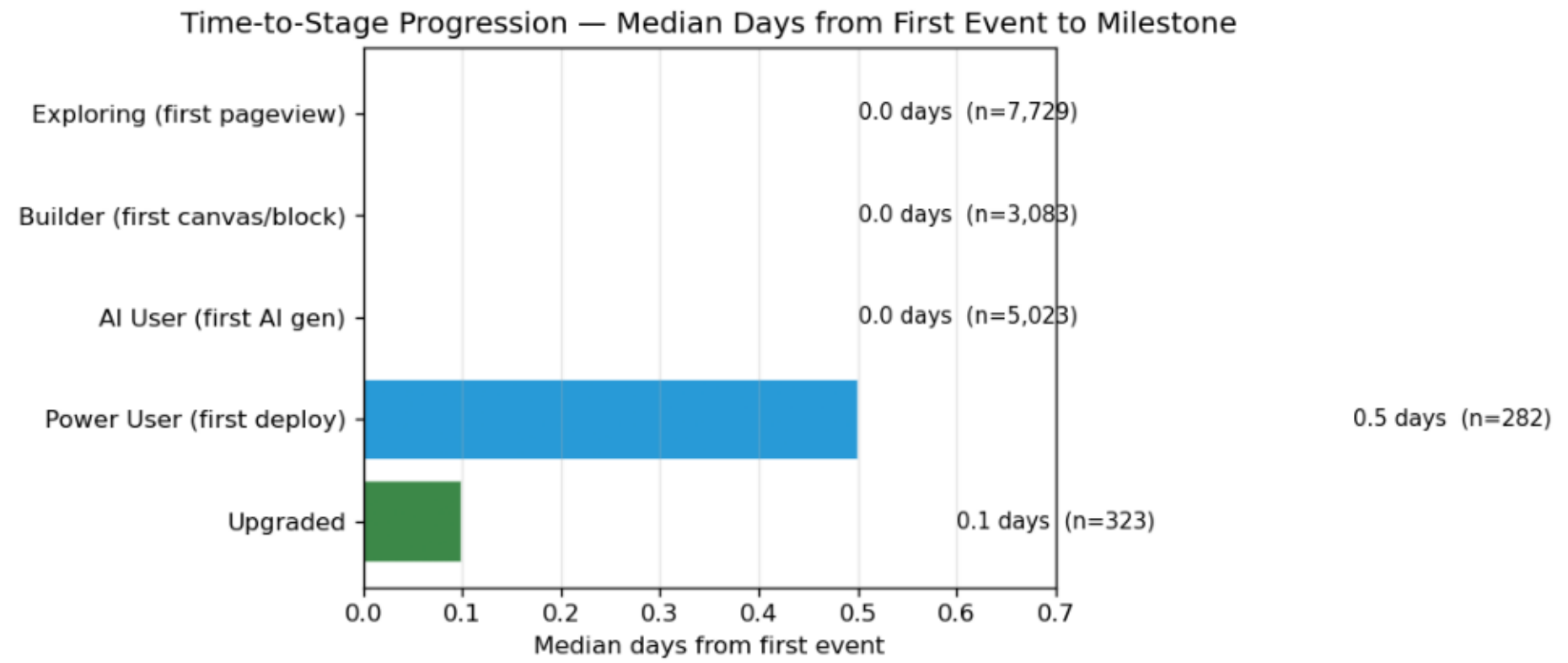
The matrix compares user stage thirty days before the dataset end with current stage, showing how users progressed, stayed still, upgraded, or became at risk.

What can be inferred

Most users do not move quickly through the funnel, so conversion work should focus on accelerating key transitions: Exploring to Builder, Builder to AI User, and AI User to Power User.

Time to Stage

Median time from first event to each funnel milestone.



What this output shows

This output shows the median number of days from first observed event to major milestones such as first pageview, first build action, first AI generation, deployment, and upgrade.

What can be inferred

Important milestones often happen very early. That implies onboarding has a narrow window to create value, and the first session or first few days are critical.

Behavioral Insights

1. AI power users drive upgrades

Upgraders averaged 86.51 AI generations versus 25.89 for non-upgraders, a 3.3x lift. Credit pressure also showed a 3.5x lift, supporting contextual upgrade nudges near AI usage limits.

2. Deployment intent is a strong signal

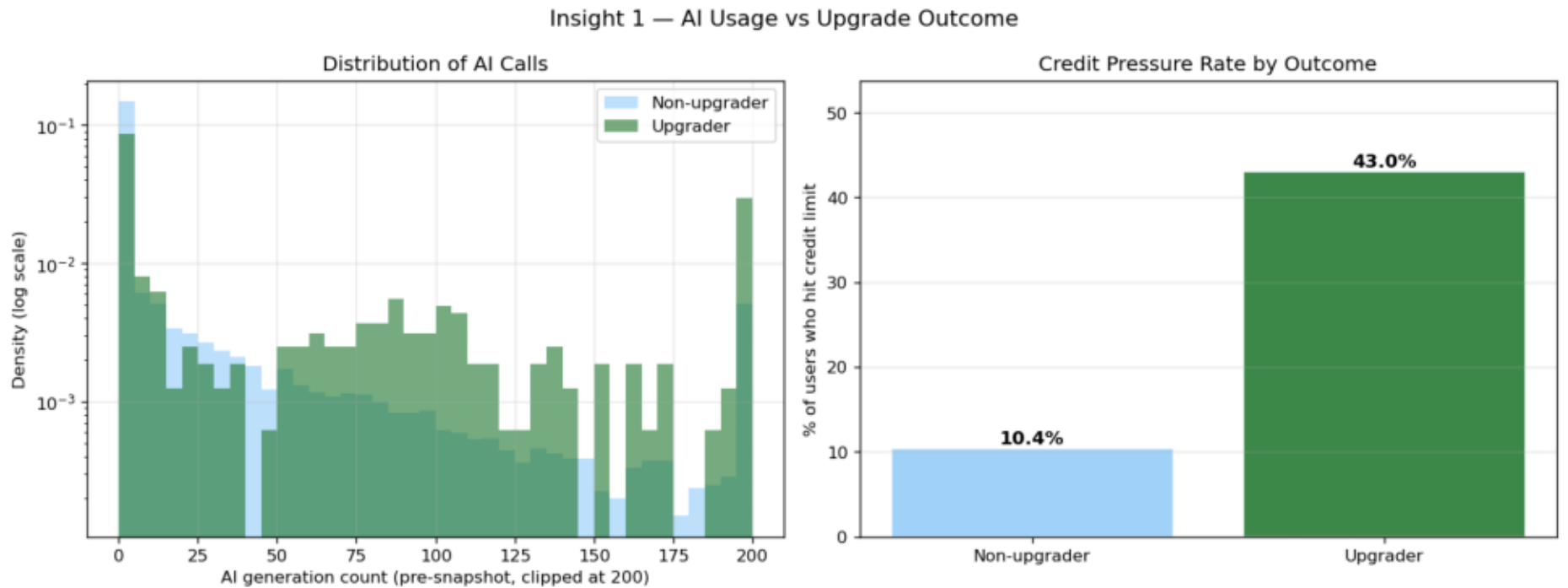
Users with deployment behavior upgraded at 9.80% versus 1.84% for non-deployers, a 5.3x lift. This suggests first deployment is a high-intent lifecycle moment.

3. Early activation matters

AI users upgraded at 4.011% overall versus 1.088% for non-AI users. The notebook recommends onboarding that gets users to their first AI generation quickly.

AI Usage and Credit Pressure

Comparison of AI and credit-pressure behavior by upgrade outcome.



What this output shows

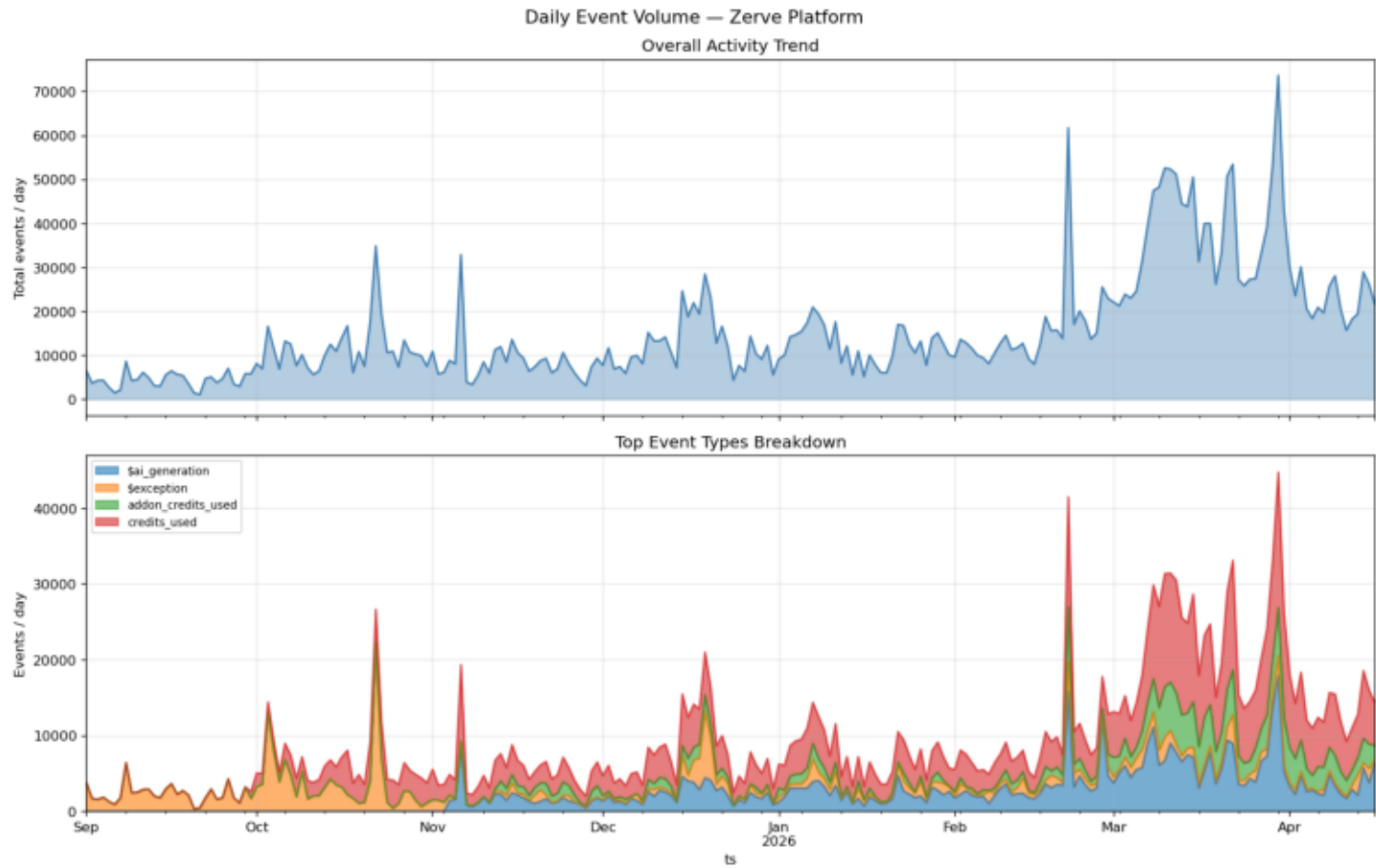
The plots compare AI generation behavior and credit-pressure behavior between users who upgraded and those who did not.

What can be inferred

Upgraders show much heavier AI usage and more credit pressure, suggesting that users who repeatedly experience AI value and resource limits are more likely to pay.

Daily Event Volume

Event-volume trend used to inspect activity patterns and data gaps.



What this output shows

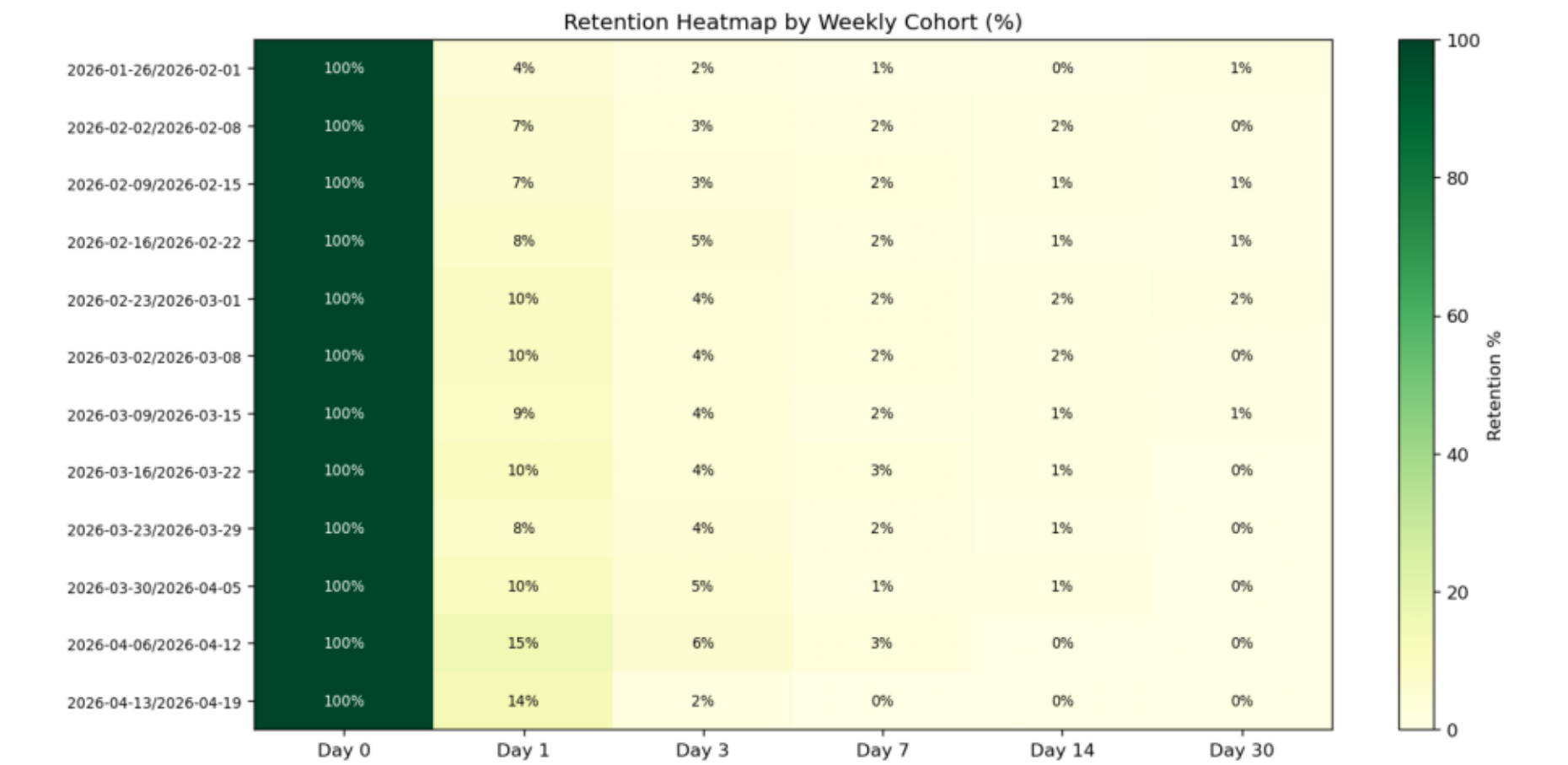
This time-series view shows aggregate event volume across the observation period and helps reveal spikes, troughs, and possible data collection gaps.

What can be inferred

Activity is not perfectly uniform over time, so future validation should consider temporal holdouts to ensure model performance generalizes beyond the observed activity pattern.

Retention Cohorts

Cohort retention view across day 0, 1, 3, 7, 14, and 30.



What this output shows

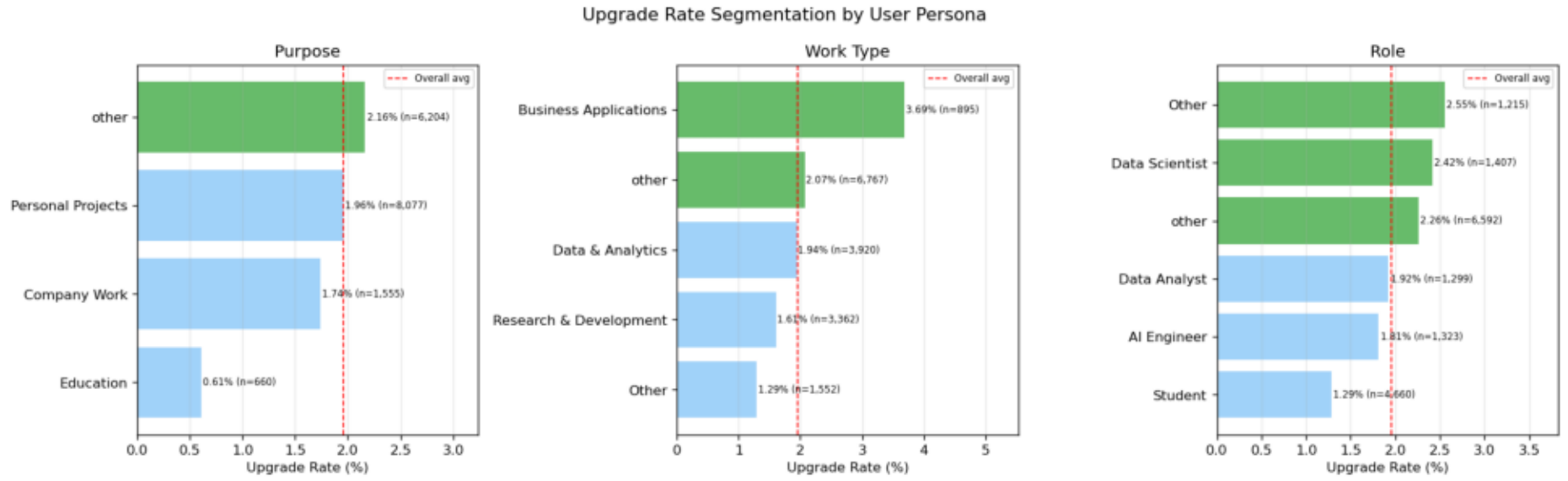
Users are grouped by signup cohort, then measured by the percentage who return on later lifecycle days.

What can be inferred

Retention appears to decline quickly after initial usage, so activation quality and early habit formation are likely important levers for downstream conversion.

Persona Segmentation

Upgrade rate and user distribution by signup persona fields.



What this output shows

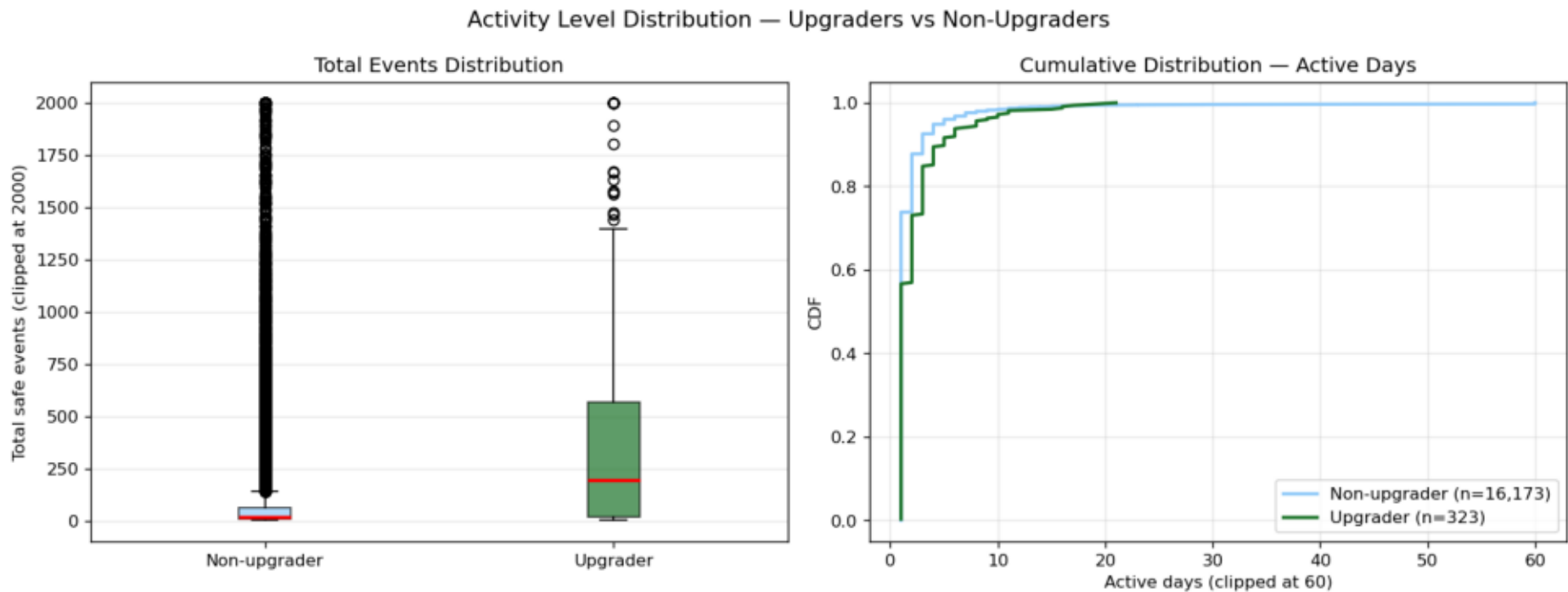
The segmentation charts compare upgrade behavior across available signup attributes such as purpose, role, source, or work type.

What can be inferred

Some personas and acquisition contexts appear more commercially qualified than others, meaning messaging and onboarding can be tailored by signup intent.

Events Per User Distribution

Distributional comparison of activity depth for upgraders and non-upgraders.



What this output shows

The distribution compares total event counts for upgraders and non-upgraders, showing how activity depth differs across outcomes.

What can be inferred

Upgraders tend to be more active, but activity alone is not enough; the model benefits from combining volume with AI usage, recency, deployment, and credit-pressure features.

Conclusion

What worked

The project combines methodological rigor with operational usefulness: leakage-safe snapshots, class-imbalance-aware evaluation, interpretable model outputs, action tiers, and a deterministic funnel that can be implemented directly.

Recommended next steps

Validate calibration on a future holdout period, monitor tier conversion rates after launch, A/B test AI quota messaging, and instrument first-deployment upgrade prompts.

Submission readiness

The notebook covers the predictive modeling challenge, funnel challenge, and behavioral insight requirements without relying on post-upgrade leakage.